

Abubakar Sadiq Shittu

📍 Knoxville, Tennessee, United States

✉ ashittu@utk.edu | 📞 865-236-7456 | 🌐 sadiq.info | 🔗 linkedin.com/in/asshittu

📅 Internship Availability: May 2027 – August 2027

🎓 EDUCATION

University of Tennessee Knoxville, United States.

01/25–05/28 (Anticipated)

Ph.D. Computer Science

GPA: 3.67/4.00

Selected Coursework: Software Engineering, Algorithms, Systems Organization, Data-Driven Security, Digital Archaeology

Advisor: Dr. Scott Ruoti

Newcastle University, Newcastle upon Tyne, United Kingdom.

09/23–09/24

M.Sc. Advanced Computer Science

GPA: 4.00/4.00

Selected Coursework: Machine Learning, Secure Software Development, Systems Security, Risk & Trust Management

Advisor: Prof. Thomas Gross

Fountain University Osogbo, Nigeria.

03/18–06/21

B.Sc. Computer Science, First Class Honours

GPA: 4.63/5.00

Advisor: Dr. Mohammed Lawal

💼 EXPERIENCE

Graduate Research Assistant

Knoxville, Tennessee, United States

Usable Security and Engineering Research Laboratory

01/25–Present

- Built and deployed a Docker-based data pipeline, using Celery to coordinate asynchronous data-collection tasks, GitHub's REST and GraphQL APIs to retrieve user and commit metadata, and PostgreSQL to store millions of records.
- Measured and analyzed existing deployments of security systems [3].
- Conducted mixed-methods user studies of security systems to identify usability, functionality, and key-management barriers to adoption and correct use [1, 2]; analyzed qualitative self-reports, log-derived behavioral data, and standardized usability metrics (ASQ and SUS), and applied inferential statistics.
- Conducting interviews with stakeholders to identify factors affecting the successful deployment of security systems.

Information Technology Helpdesk Intern

Nigeria

Nigerian National Petroleum Corporation

03/19–08/19

- Supported more than 100 users as a primary technical contact, resolving at least five issues daily and maintaining operational continuity.

📖 PUBLICATIONS

1. **Abubakar Sadiq Shittu**, Clay Shubert, John Sadik, and Scott Ruoti. "Setting up TLS authentication was hell": A Usability Study of Client Certificate Authentication. **To appear in** *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security (CCS '26)*, November 15–19, 2026, The Hague, Netherlands. Preprint: [arXiv:2604.14330](https://arxiv.org/abs/2604.14330).
2. **Abubakar Sadiq Shittu**, John Sadik, and Scott Ruoti. "A Multi-Month Study of Git Commit Signing." **To appear in** *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security (CCS '26)*, November 15–19, 2026, The Hague, Netherlands. Preprint: [arXiv:2608.29283](https://arxiv.org/abs/2608.29283).

Manuscripts Under Review

3. **Abubakar Sadiq Shittu**, John Sadik, Farzin Gholamrezae, and Scott Ruoti. "Analysis of Commit Signing on GitHub." Manuscript under review, 2026. Preprint: [arXiv:2604.14014](https://arxiv.org/abs/2604.14014).

📁 SELECTED CLASS PROJECTS

Machine Learning Pipelines

Newcastle University

- Developed classification, regression, NLP-clustering, and LSTM pipelines across FARS, bacterial-growth, news, and Raspberry Pi sensor datasets using scikit-learn, SMOTE, and Matplotlib.

Advanced RAG Pipeline

University of Tennessee, Knoxville

- Built a RAG question-answering pipeline with LangChain and Gemini by scraping university course schedules with BeautifulSoup, cleaning and storing the data as JSON, chunking documents with RecursiveCharacterTextSplitter, and indexing VertexAIEmbeddings in a vector store for semantic retrieval.

LLM-Based Assistant for Dog Care Guidance

University of Tennessee, Knoxville

- Curated a veterinary-verified dataset of 260 dog breeds and evaluated a TinyLlama-FAISS RAG pipeline against a LoRA fine-tuned LLaMA-3.2-1B model on 15 expert-verified queries.

... OTHERS

- **Technical Skills:** Python (pandas, NumPy, scikit-learn, Matplotlib, statsmodels), R, C/C++, Docker, SQL, PostgreSQL, REST/GraphQL APIs, Bash, Git, SonarQube, Wireshark, NVivo, MAXQDA, macOS, Windows, Linux/Unix.
- **Selected Awards:** Travel Grant, Summer School on Real-World Crypto and Privacy, Croatia (2024) | Chevening Scholarship, UK Govt. Fully Funded (~\$70k: tuition, stipend, travel, 2023).
- **Professional Service:** Reviewer for *Journal of Cybersecurity* (Q1 SJR, 2025).
- **Affiliations & Community:** Graduate Student Member, IEEE Computer Society | Tennessee Extension Volunteer.